Mini Project

Air Quality Index Monitoring

India — 30th April 2025

Submitted By

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CSE Department | Section S5

R-Programming Lab

SCSVMV UNIVERSITY

1. Aim

The aim of this study is to monitor and analyse the Air Quality Index (AQI) across various cities in India using real-time pollution data collected from monitoring stations. By examining AQI values and predominant pollutants, this analysis seeks to identify pollution levels, classify air quality categories, and derive insights that can support environmental decision-making and public health awareness.

2. Scenario

India's rapid urbanization and industrial growth have resulted in significant air pollution across urban and semi-urban areas. Monitoring air quality is crucial to protect public health and enforce environmental regulations. The Central Pollution Control Board (CPCB) deploys monitoring stations across Indian states to measure pollutants such as PM2.5, PM10, O3, NO2, CO, and others.

In this scenario, data has been collected from 19 cities across 11 Indian states on 30th April 2025. Each record includes the state and city of measurement, the number of active monitoring stations, the predominant pollutant parameter detected, the calculated AQI value, and the resulting AQI category (Good, Satisfactory, or Moderate).

The AQI categories used in this dataset are defined as follows:

Good (0–50): Air quality is satisfactory and poses little or no risk.

Satisfactory (51–100): Acceptable air quality; some pollutants may be a concern for a small number of sensitive individuals.

Moderate (101–200): Members of sensitive groups may experience health effects; general public is unlikely to be affected.

3. Sample Dataset

The following table presents AQI monitoring data collected from 19 Indian cities on 30th April 2025. The dataset includes the number of monitoring stations, predominant pollutant parameter, AQI index value, and air quality category for each location.

Date	State	City	No. of Stations	Predominant Parameter	AQI Value	AQI Category
30-04-2025	Assam	Nalbari	1	PM10	39	**Good**

Date	State	City	No. of Stations	Predominant Parameter	AQI Value	AQI Category
30-04-2025	Karnataka	Hubballi	1	PM10	46	Good
30-04-2025	Gujarat	Ahmedabad	8	PM2.5,PM10	110	Moderate
30-04-2025	Uttarakhand	Dehradun	1	PM10	114	Moderate
30-04-2025	Tamil Nadu	Vellore	1	PM10	51	Satisfactory
30-04-2025	Maharashtra	Ulhasnagar	1	PM10	84	Satisfactory
30-04-2025	Tamil Nadu	Chengalpattu	1	PM10	66	Satisfactory
30-04-2025	Andhra Pradesh	Tirupati	1	PM2.5	47	Good
30-04-2025	Tamil Nadu	Dindigul	1	PM10	99	Satisfactory
30-04-2025	Andhra Pradesh	Kadapa	1	O3	67	Satisfactory
30-04-2025	Maharashtra	Thane	2	PM10	58	Satisfactory
30-04-2025	Uttarakhand	Rishikesh	1	PM2.5	98	Satisfactory
30-04-2025	Chhattisgarh	Raipur	3	PM10,NO2	42	Good
30-04-2025	Tamil Nadu	Ranipet	1	PM10	39	Good
30-04-2025	Bihar	Samastipur	1	PM10	37	Good
30-04-2025	Madhya Pradesh	Jabalpur	3	PM10,O3	75	Satisfactory
30-04-2025	Bihar	Kishanganj	1	PM10	35	Good
30-04-2025	Bihar	Saharsa	1	CO	41	Good
30-04-2025	Karnataka	Yadgir	1	PM2.5	107	Moderate

```
# ============================================================
# Air Quality Index (AQI) Monitoring - R Program
# Date: 30-04-2025 | Source: CPCB India
# ============================================================
# ------ STEP 1: Dataset ------
cities <- c(
  "Nalbari", "Hubballi", "Ahmedabad", "Dehradun", "Vellore",
```

```
  "Ulhasnagar", "Chengalpattu", "Tirupati", "Dindigul", "Kadapa",
  "Thane", "Rishikesh", "Raipur", "Ranipet", "Samastipur",
  "Jabalpur", "Kishanganj", "Saharsa", "Yadgir"
)                                       # City names
aqi_values <- c(
  39, 46, 110, 114, 51,
  84, 66, 47, 99, 67,
  58, 98, 42, 39, 37,
  75, 35, 41, 107
)                                       # AQI index values
categories <- c(
  "Good", "Good", "Moderate", "Moderate", "Satisfactory",
  "Satisfactory", "Satisfactory", "Good", "Satisfactory", "Satisfactory",
  "Satisfactory", "Satisfactory", "Good", "Good", "Good",
  "Satisfactory", "Good", "Good", "Moderate"
)                                       # AQI category per city
states <- c(
  "Assam", "Karnataka", "Gujarat", "Uttarakhand", "Tamil Nadu",
  "Maharashtra", "Tamil Nadu", "Andhra Pradesh", "Tamil Nadu", "Andhra Pradesh",
  "Maharashtra", "Uttarakhand", "Chhattisgarh", "Tamil Nadu", "Bihar",
  "Madhya Pradesh", "Bihar", "Bihar", "Karnataka"
)                                       # State names
# ------ Color mapping for categories ------
```

```
bar_colors <- ifelse(categories == "Good", "#2E7D32",
            ifelse(categories == "Moderate", "#E65100", "#F9A825"))
                                        # Green=Good, Amber=Satisfactory, Orange=Moderate
# ============================================================
# GRAPH 1 - Bar Plot: AQI Values by City
# ============================================================
par(mar = c(8, 5, 4, 2))                    # Set margins for rotated labels
barplot(
  aqi_values,                               # Heights of bars
  names.arg = cities,                       # City names on X-axis
  col       = bar_colors,                   # Bar colors by category
  border    = "white",                      # White border between bars
  las       = 2,                            # Rotate X-axis labels vertically
  cex.names = 0.75,                         # Reduce city name font size
  ylim      = c(0, 130),                    # Y-axis range
  main      = "AQI Values by City (30-Apr-2025)",  # Chart title
  xlab      = "",                           # No X label (labels rotated)
  ylab      = "AQI Value (Indices)",        # Y-axis label
  cex.main  = 1.2                           # Title font size
)
abline(h = 50,  col = "green",  lty = 2, lwd = 1.5)   # Good boundary line
abline(h = 100, col = "orange", lty = 2, lwd = 1.5)   # Satisfactory boundary line
```

```
text(                                   # Add value labels on top of bars
  x      = seq(0.7, by = 1.2, length.out = length(aqi_values)),
  y      = aqi_values + 3,                      # Slightly above each bar
  labels = aqi_values,                          # Display AQI value
  cex    = 0.75,                              # Font size
  col    = "black"                            # Label color
)
legend(                                       # Add legend
  "topright",
  legend = c("Good", "Satisfactory", "Moderate"),
  fill   = c("#2E7D32", "#F9A825", "#E65100"),
  title  = "AQI Category",
  cex    = 0.85,
  bty    = "n"                                # No legend box border
)
# ============================================================
# GRAPH 2 - Pie Chart: AQI Category Distribution
# ============================================================
good_count         <- sum(categories == "Good")          # Count Good cities
satisfactory_count <- sum(categories == "Satisfactory")  # Count Satisfactory cities
moderate_count     <- sum(categories == "Moderate")      # Count Moderate cities
pie_values <- c(good_count, satisfactory_count, moderate_count)   # Slice sizes
pie_labels <- c(                                        # Slice labels with count
```

```
  paste0("Good\n", good_count, " cities"),
  paste0("Satisfactory\n", satisfactory_count, " cities"),
  paste0("Moderate\n", moderate_count, " cities")
)
pie_colors <- c("#2E7D32", "#F9A825", "#E65100")          # Slice colors
pie(
  pie_values,                                  # Slice sizes
  labels  = pie_labels,                          # Labels on slices
  col     = pie_colors,                          # Slice colors
  main    = "AQI Category Distribution\n19 Cities | 30-Apr-2025",  # Title
  cex     = 0.9                                  # Label font size
)
legend(                                          # Add legend
  "bottomright",
  legend = c("Good", "Satisfactory", "Moderate"),
  fill   = pie_colors,
  title  = "AQI Category",
  cex    = 0.85,
  bty    = "n"
)
# ============================================================
# GRAPH 3 - Bar Plot: State-wise Average AQI
# ============================================================
```

```
state_avg <- tapply(aqi_values, states, mean)      # Calculate average AQI per state
state_avg <- round(state_avg, 1)                   # Round to 1 decimal
state_avg <- sort(state_avg)                       # Sort ascending
par(mar = c(9, 5, 4, 2))                           # Margins for rotated labels
barplot(
  state_avg,                                       # Average AQI values
  names.arg = names(state_avg),                    # State names on X-axis
  col       = "#2E75B6",                           # Blue bars
  border    = "white",
  las       = 2,                                   # Rotate labels
  cex.names = 0.8,
  ylim      = c(0, 130),
  main      = "State-wise Average AQI Value\n30-Apr-2025",
  xlab      = "",
  ylab      = "Average AQI (Indices)",
  cex.main  = 1.2
)
text(                                              # Value labels on bars
  x      = seq(0.7, by = 1.2, length.out = length(state_avg)),
  y      = state_avg + 3,
  labels = state_avg,
  cex    = 0.8,
  col    = "black"
```

```
)
# ============================================================
# GRAPH 4 - Horizontal Bar Plot: City AQI Ranking
# ============================================================
sorted_order  <- order(aqi_values)                # Sort cities by AQI ascending
sorted_aqi    <- aqi_values[sorted_order]         # Sorted AQI values
sorted_cities <- cities[sorted_order]             # Sorted city names
sorted_colors <- bar_colors[sorted_order]         # Sorted colors
par(mar = c(4, 8, 4, 3))                          # Wide left margin for city names
barplot(
  sorted_aqi,                                     # AQI values (sorted)
  names.arg = sorted_cities,                      # City names on Y-axis
  col       = sorted_colors,                      # Colors by category
  border    = "white",
  horiz     = TRUE,                               # Horizontal bars
  las       = 1,                                  # Horizontal labels
  cex.names = 0.75,
  xlim      = c(0, 135),
  main      = "City-wise AQI Ranking (Low to High)\n30-Apr-2025",
  xlab      = "AQI Value (Indices)",
  ylab      = "",
  cex.main  = 1.2
)
```

```
text(                                   # Value labels at end of bars
  x      = sorted_aqi + 2,
  y      = seq(0.7, by = 1.2, length.out = length(sorted_aqi)),
  labels = sorted_aqi,
  cex    = 0.75,
  col    = "black"
)
legend(                                 # Legend
  "bottomright",
  legend = c("Good", "Satisfactory", "Moderate"),
  fill   = c("#2E7D32", "#F9A825", "#E65100"),
  title  = "AQI Category",
  cex    = 0.85,
  bty    = "n"
)
```

Overall Observation

*The data was collected from **19 cities in India** on **30th April 2025**.*

Air Quality Summary

Out of 19 cities:

- ***8 cities** had **Good** air quality*
- ***8 cities** had **Satisfactory** air quality*
- ***3 cities** had **Moderate** air quality*

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```python
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches

# ============================================================
# AQI Monitoring - Graphs using Python (Pandas & Matplotlib)
# ============================================================

# ------ STEP 1: Dataset ------

cities = [
  "Nalbari", "Hubballi", "Ahmedabad", "Dehradun", "Vellore",
  "Ulhasnagar", "Chengalpattu", "Tirupati", "Dindigul", "Kadapa",
  "Thane", "Rishikesh", "Raipur", "Ranipet", "Samastipur",
  "Jabalpur", "Kishanganj", "Saharsa", "Yadgir"
]                                                    # City names

aqi_values = [
  39, 46, 110, 114, 51,
  84, 66, 47, 99, 67,
  58, 98, 42, 39, 37,
  75, 35, 41, 107
]                                                    # AQI index values

categories = [
  "Good", "Good", "Moderate", "Moderate", "Satisfactory",
  "Satisfactory", "Satisfactory", "Good", "Satisfactory", "Satisfactory",
```

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```
    "Assam", "Karnataka", "Gujarat", "Uttarakhand", "Tamil Nadu",
    "Maharashtra", "Tamil Nadu", "Andhra Pradesh", "Tamil Nadu", "Andhra Pradesh",
    "Maharashtra", "Uttarakhand", "Chhattisgarh", "Tamil Nadu", "Bihar",
    "Madhya Pradesh", "Bihar", "Bihar", "Karnataka"
]                                                   # State names

# Create a DataFrame
df = pd.DataFrame({
    'city': cities,
    'aqi_value': aqi_values,
    'category': categories,
    'state': states
})

# ------ Color mapping for categories ------

color_map = {
    "Good": "#2E7D32",        # Green
    "Moderate": "#E65100",    # Orange
    "Satisfactory": "#F9A825" # Yellow/Amber
}

df['color'] = df['category'].map(color_map)

# ============================================================
# GRAPH 1 - Bar Plot: AQI Values by City
# ============================================================
```

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```python
plt.xlabel("") # No X label, as labels are rotated
plt.ylabel("AQI Value (Indices)", fontsize=12)
plt.xticks(rotation=90, ha='right', fontsize=9)
plt.yticks(fontsize=10)

# Add boundary lines
plt.axhline(y=50, color='green', linestyle='--', linewidth=1.5, label='Good Boundary')
plt.axhline(y=100, color='orange', linestyle='--', linewidth=1.5, label='Satisfactory Boundary')

# Add value labels on top of bars
for i, val in enumerate(df['aqi_value']):
    plt.text(i, val + 3, str(val), ha='center', va='bottom', fontsize=9, color='black')

# Add legend
legend_patches = [mpatches.Patch(color=color_map[cat], label=cat) for cat in color_map]
plt.legend(handles=legend_patches, title="AQI Category", loc="upper right", fontsize=10, title_fontsize=11)

plt.tight_layout() # Adjust layout to prevent labels from being cut off
plt.show()

# ============================================================
# GRAPH 2 - Pie Chart: AQI Category Distribution
# ============================================================

category_counts = df['category'].value_counts().reindex(["Good", "Satisfactory", "Moderate"])
pie_labels = [f"{cat}\n{count} cities" for cat, count in category_counts.items()]
pie_colors = [color_map[cat] for cat in category_counts.index]
```

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[7] 1s

```python
legend_patches_pie = [mpatches.Patch(color=color_map[cat], label=cat) for cat in color_map]
plt.legend(handles=legend_patches_pie, title="AQI Category", loc="lower right", fontsize=10, t

plt.tight_layout()
plt.show()

# ============================================================
# GRAPH 3 - Bar Plot: State-wise Average AQI
# ============================================================

state_avg = df.groupby('state')['aqi_value'].mean().round(1).sort_values()

plt.figure(figsize=(12, 7)) # Adjust figure size
plt.bar(state_avg.index, state_avg.values, color="#2E75B6")

plt.ylim(0, 130)
plt.title("State-wise Average AQI Value\n30-Apr-2025", fontsize=16)
plt.xlabel("")
plt.ylabel("Average AQI (Indices)", fontsize=12)
plt.xticks(rotation=90, ha='right', fontsize=10)
plt.yticks(fontsize=10)

# Add value labels on bars
for i, val in enumerate(state_avg.values):
    plt.text(i, val + 3, str(val), ha='center', va='bottom', fontsize=9, color='black')

plt.tight_layout()
plt.show()
```

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[7]

```python
# Add value labels on bars
for i, val in enumerate(state_avg.values):
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plt.tight_layout()
plt.show()

# ============================================================
# GRAPH 4 - Horizontal Bar Plot: City AQI Ranking
# ============================================================

df_sorted = df.sort_values(by='aqi_value', ascending=True)

plt.figure(figsize=(12, 9)) # Adjust figure size for horizontal bars
plt.barh(df_sorted['city'], df_sorted['aqi_value'], color=df_sorted['color'])

plt.xlim(0, 135) # x-axis range for horizontal bars
plt.title("City-wise AQI Ranking (Low to High)\n30-Apr-2025", fontsize=16)
plt.xlabel("AQI Value (Indices)", fontsize=12)
plt.ylabel("") # No Y label
plt.yticks(fontsize=9)
plt.xticks(fontsize=10)

# Add value labels at end of bars
for i, val in enumerate(df_sorted['aqi_value']):
    plt.text(val + 2, i, str(val), ha='left', va='center', fontsize=9, color='black')

# Add legend
```

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[7]

```python
# GRAPH 4 - Horizontal Bar Plot: City AQI Ranking
# ==========================================================

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plt.ylabel("") # No Y label
plt.yticks(fontsize=9)
plt.xticks(fontsize=10)

# Add value labels at end of bars
for i, val in enumerate(df_sorted['aqi_value']):
    plt.text(val + 2, i, str(val), ha='left', va='center', fontsize=9, color='black')

# Add legend
legend_patches_horiz = [mpatches.Patch(color=color_map[cat], label=cat) for cat in color_map]
plt.legend(handles=legend_patches_horiz, title="AQI Category", loc="lower right", fontsize=10, title_fontsize=11)

plt.tight_layout()
plt.show()
```

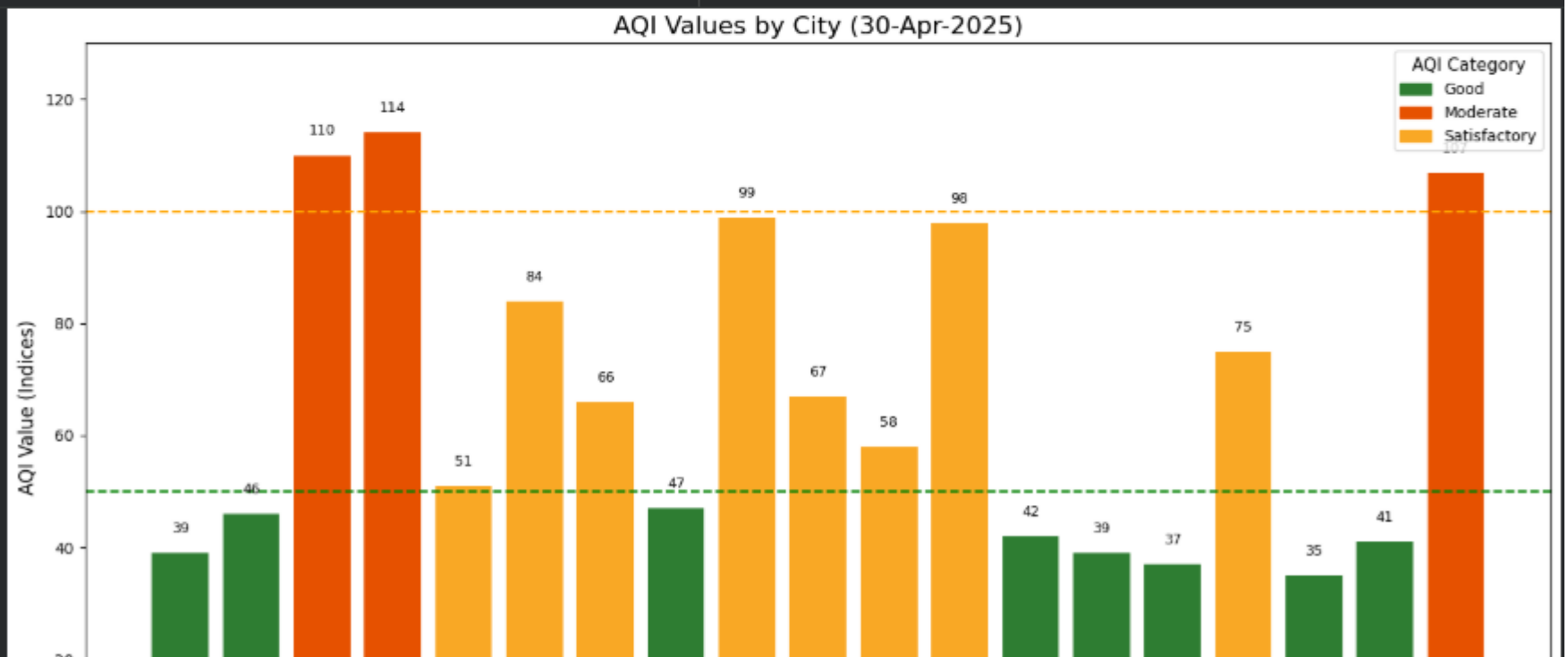

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# GRAPH 4 - Horizontal Bar Plot: City AQI Ranking
# ==============================================

df_sorted = df.sort_values(by='aqi_value', ascending=True)

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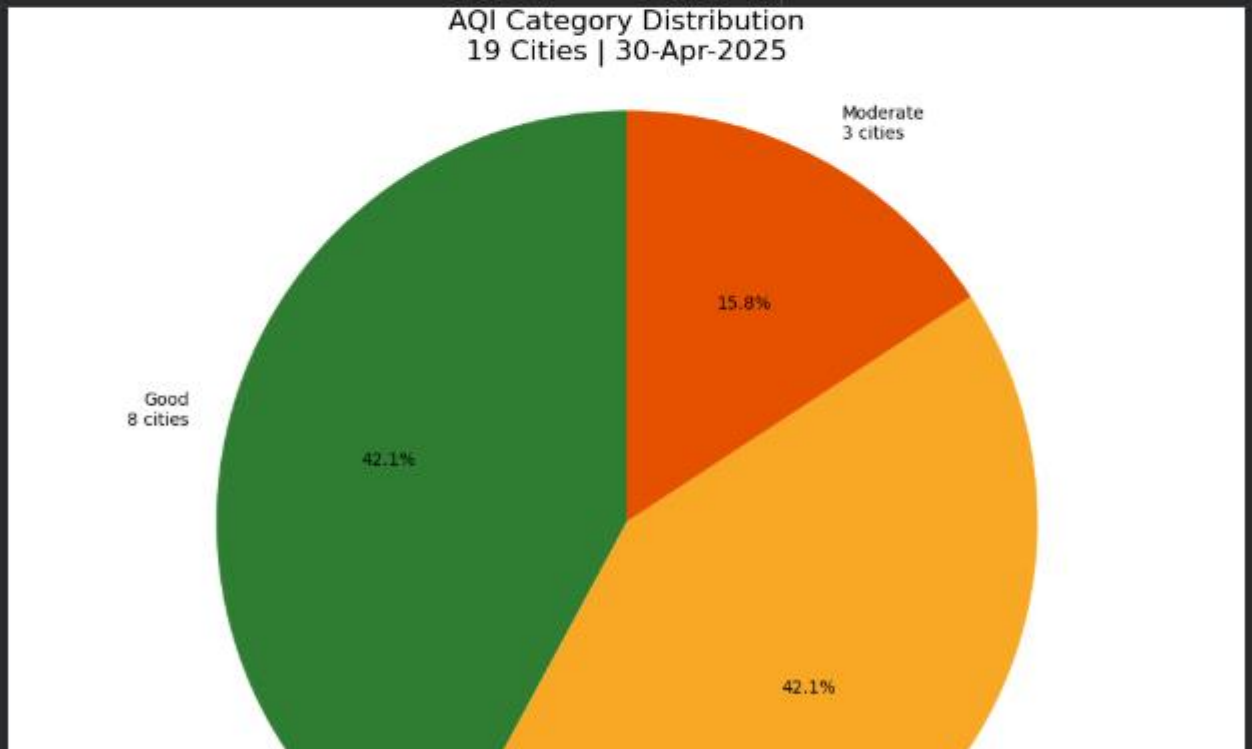

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```python
# GRAPH 4 - Horizontal Bar Plot: City AQI Ranking
# ==========================================================

df_sorted = df.sort_values(by='aqi_value', ascending=True)

plt.figure(figsize=(12, 9)) # Adjust figure size for horizontal bars
plt.barh(df_sorted['city'], df_sorted['aqi_value'], color=df_sorted['color'])

plt.xlim(0, 135) # x-axis range for horizontal bars
plt.title("City-wise AQI Ranking (Low to High)\n30-Apr-2025", fontsize=16)
plt.xlabel("AQI Value (Indices)", fontsize=12)
plt.ylabel("") # No Y label
plt.yticks(fontsize=9)
plt.xticks(fontsize=10)

# Add value labels at end of bars
for i, val in enumerate(df_sorted['aqi_value']):
    plt.text(val + 2, i, str(val), ha='left', va='center', fontsize=9, color='black')

# Add legend
legend_patches_horiz = [mpatches.Patch(color=color_map[cat], label=cat) for cat in color_map]
plt.legend(handles=legend_patches_horiz, title="AQI Category", loc="lower right", fontsize=10, title_fontsize=11)

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plt.show()
```

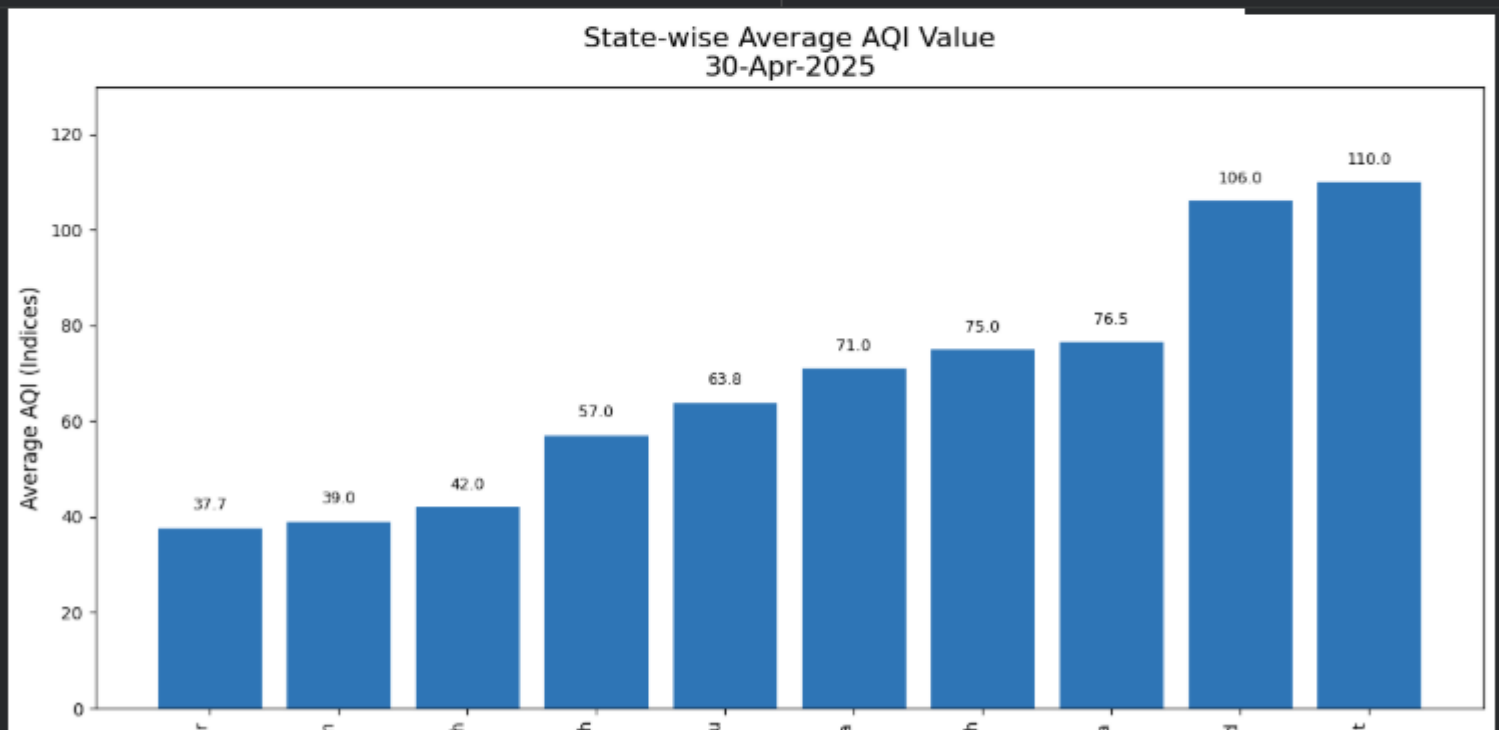

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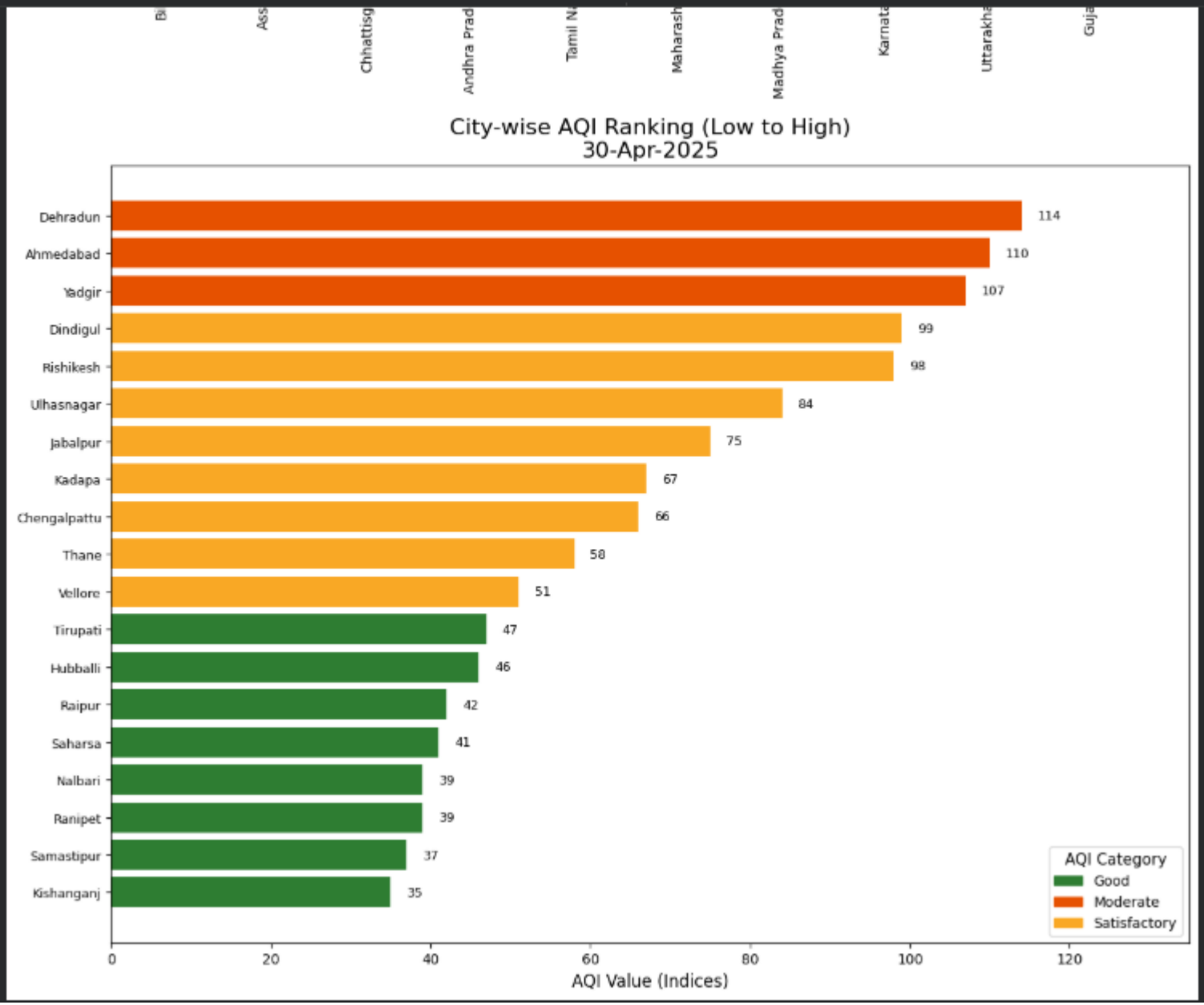

Source: Central Pollution Control Board (CPCB), India — 30 April 2025